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Author(s): L. Chwistek

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A FORMAL PROOF OF GÖDEL'S THEOREM

L. CHWISTEK

I. The present paper contains a formal proof of the following theorem of the elementary system $\models [10]c$:¹

$$\models [10] \supset \vdash_0 (32) \sim \vdash_2 (54) = .0(5) .1(5) [5] \\ \vdash_0 (32) = .0(3) .1(3) [3].$$

This theorem means that if there were a theorem of $\vdash_0 (32) 0$ stating that there is no contradiction in $\vdash_2 (54) 2$, we would have a contradiction in $\vdash_2 (54) 2$. (This is in accordance with the second theorem of Gödel.)

Our proof is based on the calculus of the system $\models [10]c$ and of the meta-system $\vdash_0 (32) 0$.

Note that we could take²

$$\Lambda (.0(E) .1(.1(L)) .0(.1(L)) E) [L] \vee = Z \models [.1(L) .0(L)] \\ \text{Expr } [.1(L)] .0(.1(L)) [L] = Z \models [.1(L) .0(L)] \\ \text{Expr } \Omega E .1(L) L [L]$$

instead of $Ax E .1(L) (.1(L) .0(L) L) Z$. We would then have to do with the simple systems and metasystems of Hetper, which do not contain propositional variables or the logico-semantical axiom.³ Our method applies equally to the systems and metasystems of *New Foundation* and to the simple systems and metasystems of Hetper.

Our proof can be considerably simplified by using a recent result of Hetper concerning *ancestral functions*.⁴ Hetper introduces the following abbreviations:

$\epsilon_L X I$	is an abbreviation of	$\exists [.0(L) L] \bar{s}_L \bar{t}_L \Lambda = I * \bar{s}_L \bar{t}_L [.0(L)] \\ \{ \bar{t}_L * \bar{s}_L X \} [.0(L)].$
$\text{Princ}_L (A) Z$	"	$\exists [.0(L) L] \bar{z}_L \Lambda = Z * \bar{z}_L .1(L) [.0(L)] A.$
$\text{Der}_L (B) X Y Z$	"	$\exists [.0(L) L] \bar{p}_L \bar{q}_L \bar{x}_L \bar{y}_L \bar{z}_L \\ \Lambda = X * \bar{x}_L \bar{p}_L [.0(L)] \Lambda = Y * \bar{y}_L \bar{q}_L [.0(L)] \\ \Lambda = Z * \bar{z}_L * .1(L) \bar{p}_L \bar{q}_L [.0(L)] B.$
$\text{Anc}_L (A B) Z$	"	$\exists [.0(L) L] \bar{k}_{1L} \bar{t}_{1L} \Lambda \epsilon_L * Z \bar{k}_{1L} \bar{t}_{1L} \\ \Pi [.0(L) L] \bar{z}_{1L} \supset \epsilon_L \bar{z}_{1L} \bar{t}_{1L} \vee \text{Princ}_L (A) \bar{z}_{1L} \\ \exists [.0(L) L] \bar{x}_{1L} \bar{y}_{1L} \Lambda \epsilon_L \bar{x}_{1L} \bar{t}_{1L} \Lambda \epsilon_L \bar{y}_{1L} \bar{t}_{1L} \\ \text{Der}_L (B) \bar{x}_{1L} \bar{y}_{1L} \bar{z}_{1L}.$

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¹ Cf. L. Chwistek and W. Hetper, *New foundation of formal metamathematics*, this JOURNAL, vol. 3 (1938), p. 36.

² Loc. cit., p. 19.

³ Cf. W. Hetper, *Simple systems, metasystems, and generalized metasystems* (to appear).

⁴ Cf. W. Hetper, *Relacje ancestralne w systemie semantyki*, *Archiwum Towarzystwa Naukowego we Lwowie*, vol. 9, no. 6.

If $A(z_{1c})$, $B(x_{1c} y_{1c} z_{1c})$ are propositions, the expression $\text{Anc}_0(A(\bar{z}_0) B(\bar{x}_0 \bar{y}_0 \bar{z}_0)) a_{1c}$ will be called an *ancestral function* (we prove without difficulty that this expression is a proposition). The expressions $A(\bar{z}_0)$, $B(\bar{x}_0 \bar{y}_0 \bar{z}_0)$ will be called respectively the *principal term* and the *term of derivation* of this function.

Let us now suppose that $\Phi(a_{1c})$ is an ancestral function and $A(\bar{z}_0)$, $B(\bar{x}_0 \bar{y}_0 \bar{z}_0)$ are its principal term and its term of derivation. In accordance with the theory of Hetper, if

$$\models [10] \supset A(z_{1c}) \Psi(z_{1c}),$$

$$\models [10] \supset \Lambda \Psi(x_{1c}) \Lambda \Psi(y_{1c}) B(x_{1c} y_{1c} z_{1c}) \Psi(z_{1c})$$

are theorems, then $\models [10] \supset \Phi(z_{1c}) \Psi(z_{1c})$ is a theorem.

This statement will be called the *ancestral pattern*.

II. Let us begin with some abbreviations which will enable us to get a very simple symbolic language. We assume the following directions:⁵

$Ax E(L) Z$	is an abbreviation of	$Ax E.2(L) (.2(L).1(L).0(L))$
		$\models [.2(L).1(L)] Z.$
$Ax_L Z$	"	$\exists [.0(L)] \bar{e}_L Ax \bar{e}_L(L) Z.$
$DEFGHIK(L) XYZ$	"	$DEFGHIK(.2(L).1(L).0(L))$
		$\models [.2(L).1(L)] X$
		$\models [.2(L).1(L)] Y$
		$\models [.2(L).1(L)] Z.$
$D(L) XYZ$	"	$D_L(.2(L)) \models [.2(L).1(L)] X$
		$\models [.2(L).1(L)] Y$
		$\models [.2(L).1(L)] Z.$
$\vdash_L E$	"	$\vdash_L (.2(L).1(L)) E.$
$\text{Im}_L EF$	"	$(E.0(L).2(L)F)[L].$
$S_L E$	"	$\exists [.0(L)L] \bar{s}_{2L} \wedge \text{Im}_{.0(L)} E \bar{s}_{2L} \vdash_L \bar{s}_{2L}.$
$S'_L (Ax)_L Z$	"	$\exists [.0(L)L] \bar{z}_{2L} \wedge \text{Im}_{.0(L)} Z \bar{z}_{2L}$
		$\vdash_L Ax_{.1(L)} \bar{z}_{2L}.$
$S'_L (D)_L XYZ$	"	$\exists [.0(L)L] \bar{x}_{2L} \bar{y}_{2L} \bar{z}_{2L}$
		$\wedge \text{Im}_{.0(L)} **XYZ** \bar{x}_{2L} \bar{y}_{2L} \bar{z}_{2L}$
		$\vdash_L D_{.1(L)} \bar{x}_{2L} \bar{y}_{2L} \bar{z}_{2L}.$
$S_{LM} E$	"	$\exists [.0(L)L] \bar{s}_{2L} \wedge \text{Im}_{.0(L)} E \bar{s}_{2L}$
		$\vdash_L \vdash_M \bar{s}_{2L}.$

Our argument will be based on the following theorem:

⁵ Cf. *New foundation*, p. 35.

$$\models [1\ 0] \supset \vdash_0 u_{1c} S_{02} u_{1c}. \quad (a)$$

We shall prove this theorem using the ancestral pattern. To do it we prove first the following theorem, by a simple application of the principle of induction:

$$\models [1\ 0] \equiv \vdash_0 u_{1c} \text{Anc}_0 (Ax_0 (3) \bar{x}_0 D_0 (3) \bar{x}_0 \bar{y}_0 \bar{z}_0) \models [3\ 2] u_{1c}.$$

Note that Hetper dispenses with the use of this theorem, taking $\vdash_L E$ as an abbreviation of $\text{Anc}_L (Ax_L (.2 (L)) \bar{x}_L D_L (.2 (L)) \bar{x}_L \bar{y}_L \bar{z}_L) \models [.2 (L) .1 (L)] E$.

Now, $\vdash_0 Z$ being equivalent to an ancestral function, we can use it in our ancestral pattern. We take in this pattern:

$$\begin{aligned} \vdash_0 Z & \text{ for } \Phi(Z), \\ Ax_0 Z & \text{ " } A(Z), \\ D_0 X Y Z & \text{ " } B(X Y Z), \\ S_{02} Z & \text{ " } \Psi(Z). \end{aligned}$$

Then we have to prove:

$$\models [1\ 0] \supset Ax_0 u_{1c} S_{02} u_{1c}. \quad (b)$$

$$\models [1\ 0] \supset \wedge S_{02} x_{1c} \wedge S_{02} y_{1c} D_0 x_{1c} y_{1c} z_{1c} S_{02} z_{1c}. \quad (c)$$

To do it, we have first to prove:

$$\models [1\ 0] \supset Ax\ e_{1c} (0) z_{1c} S_0 Ax\ e_{1c} (0) z_{1c}. \quad (d)$$

$$\begin{aligned} \models [1\ 0] \supset D_{e_{1c} f_{1c} g_{1c} h_{1c} i_{1c} k_{1c}} (0) x_{1c} y_{1c} z_{1c} \\ S_0 D_{e_{1c} f_{1c} g_{1c} h_{1c} i_{1c} k_{1c}} (0) x_{1c} y_{1c} z_{1c}. \quad (e) \end{aligned}$$

We begin with the proof of (c) and of (d).

III. We have by the logical calculus of $\models [1\ 0]c$ and by the direction of expressions of $\vdash_0 0$:

$$(1) \quad \models [1\ 0] S_0 \text{Expr} [1] a_{1c} \text{ is a theorem.}$$

By the directions of expressions and of pattern of substitution we have:

$$(2) \quad \models [1\ 0] S_0 \text{Prop} (a_{1c} b_{1c} c_{1c} d_{1c}) [1] \text{ is a theorem.}$$

Then we get by the direction of propositions:

$$(3) \quad \models [1\ 0] S_0 \text{Prop} \sim (a_{1c} b_{1c} c_{1c} d_{1c}) [1] \text{ is a theorem.}$$

These theorems enable us to prove by the principle of induction:

$$\models [1\ 0] \supset = a_{1c} b_{1c} [1] S_0 = a_{1c} b_{1c} [1] \text{ is a theorem,}$$

$$\models [1\ 0] \supset \sim = a_{1c} b_{1c} [1] S_0 \sim = a_{1c} b_{1c} [1] \text{ is a theorem.}$$

Suppose now we have proved,

$$\models [1\ 0] S_0 \text{Prop } E, \quad \models [1\ 0] S_0 \text{Prop } F;$$

then we can prove:

$$\begin{aligned} \models [10] S_0 \supset E \vee EF, & \quad \models [10] S_0 \supset F \vee EF, \\ \models [10] \supset S_0 E S_0 \vee EF, & \quad \models [10] \supset S_0 F S_0 \vee EF, \\ \models [10] \supset \supset E S_0 E \supset E S_0 \vee EF, & \quad \models [10] \supset \supset F S_0 F \supset F S_0 \vee EF. \end{aligned}$$

Then we have by the logical calculus,

$$\models [10] \supset \wedge \supset E S_0 E \supset F S_0 F \supset \vee EF S_0 \vee EF.$$

Then we get, $\models [10] \supset \wedge \supset E S_0 E \supset F S_0 F \supset \wedge EF S_0 \wedge EF$.

Using this method, we get by the principle of induction:

$$\models [10] \supset (a_{1c} b_{1c} c_{1c} d_{1c}) [1] S_0 (a_{1c} b_{1c} c_{1c} d_{1c}) [1] \text{ is a theorem,}$$

$$\models [10] \supset \sim (a_{1c} b_{1c} c_{1c} d_{1c}) [1] S_0 \sim (a_{1c} b_{1c} c_{1c} d_{1c}) [1] \text{ is a theorem.}$$

Now, using the same method, we prove step by step the theorems (d), (e). Thus we have:

$$(4) \quad \models [10] \supset Ax e_{1c} (0) z_{1c} S_0 Ax e_{1c} (0) z_{1c} \text{ is a theorem,}$$

$$(5) \quad \models [10] \supset D e_{1c} f_{1c} g_{1c} h_{1c} i_{1c} k_{1c} (0) x_{1c} y_{1c} z_{1c} \\ S_0 D e_{1c} f_{1c} g_{1c} h_{1c} i_{1c} k_{1c} (0) x_{1c} y_{1c} z_{1c} \text{ is a theorem.}$$

Note that these theorems could not be proved if $Ax e_{1c} (0) z_{1c}$ and $D e_{1c} f_{1c} g_{1c} h_{1c} i_{1c} k_{1c} (0) x_{1c} y_{1c} z_{1c}$ were replaced by arbitrary functions.

Now we pass to the proof of the theorems (b), (c).

IV. Let us assume the following direction:

$$\bar{D} E F G \text{ is an abbreviation of } \supset D E F G h_{131} i_{131} k_{131} (2) x_{131} y_{131} z_{131} \\ D_2 x_{131} y_{131} z_{131}.$$

We have by the logical calculus of $\vdash_0 0$:

$$\begin{aligned} (6) \quad & \models [10] \vdash_0 \bar{D} e_{131} f_{131} g_{131} \text{ is a theorem.} \\ & \models [10] \vdash_0 \text{Prop } \bar{D} e_{131} f_{131} g_{131} \text{ is a theorem.} \\ & \models [10] \vdash_0 \text{Prop Prop } \bar{D} e_{131} f_{131} g_{131} \text{ is a theorem.} \end{aligned}$$

Now we get by the semantical calculus of $\models [10] c$:

$$\begin{aligned} (7) \quad & \models [10] \supset \text{Im}_1 a_{1c} a_{11c} \sim = a_{11c} .0 (1) [1] \text{ is a theorem.} \\ & \models [10] \supset \text{Im}_1 a_{1c} a_{11c} \sim = a_{11c} .0 (2) [1] \text{ is a theorem.} \\ & \models [10] \supset \text{Im}_1 a_{1c} a_{11c} \sim \{ a_{11c} I (1) \} [1] \text{ is a theorem.} \\ & \models [10] \supset \text{Im}_1 a_{1c} a_{11c} \sim \{ a_{11c} I (2) \} [1] \text{ is a theorem.} \end{aligned}$$

These theorems enable us to prove, step by step, using the semantical calculus of $\models [10] c$:

$$(8) \quad \models [10] \sim \{ \bar{D} e_{131} f_{131} g_{131} e_{132} \} [1] \text{ is a theorem.}$$

Then we get by the logical calculus of $\vdash_0 0$:

$$(9) \quad \models [10] \vdash_0 \Pi [32] e_{132} \bar{D} e_{132} f_{131} g_{131} \text{ is a theorem.}$$

Now we have by the direction of deduction of $\vdash_0 0$ and by (1) and (6),

$$\models [10] \supset \text{Im}_1 e_{1c} e_{11c} \vdash_0 \supset \Pi [32] e_{132} \bar{D} e_{132} f_{131} g_{131} \bar{D} e_{11c} f_{131} g_{131} \text{ is a theorem.}$$

Then we get by the direction of *modus ponens* of $\vdash_0 0$ in consequence of (9),

$$(10a) \quad \models [10] \supset \text{Im}_1 e_{1c} e_{11c} \vdash_0 \bar{D} e_{11c} f_{131} g_{131} \text{ is a theorem.}$$

The same method enables us to prove,

$$(10b) \quad \models [10] \supset \text{Im}_1 e_{1c} e_{11c} \vdash_0 \text{Prop } \bar{D} e_{11c} f_{131} g_{131} \text{ is a theorem.}$$

Then we have by the direction of propositions of $\vdash_0 0$,

$$(10c) \quad \models [10] \supset \text{Im}_1 e_{1c} e_{11c} \vdash_0 \text{Prop Prop } \bar{D} e_{11c} f_{131} g_{131} \text{ is a theorem.}$$

Now let us remark that the theorems (7) enable us to prove by the elementary semantical calculus:

$$(11) \quad \models [10] \supset \text{Im}_1 e_{1c} e_{11c} \sim \{ \bar{D} e_{11c} f_{131} g_{131} f_{132} \} [1] \text{ is a theorem.}$$

Then we prove using (10a), (10b), (10c), and (11) instead of (6) and (8),

$$\models [10] \supset \wedge \text{Im}_1 e_{1c} e_{11c} \text{Im}_1 f_{1c} f_{11c} \vdash_0 \bar{D} e_{11c} f_{11c} g_{131} \text{ is a theorem.}$$

Now we introduce the following directions:

Hip₁ is an abbreviation of $\text{Im}_1 ** x_{1c} y_{1c} z_{1c} ** x_{11c} y_{11c} z_{11c}$.

Hip₂ “ “ “ “ $\text{Im}_1 ***** e_{1c} f_{1c} g_{1c} h_{1c} i_{1c} k_{1c} ***** e_{11c} f_{11c} g_{11c} h_{11c} i_{11c} k_{11c}$.

Hip “ “ “ “ $\wedge \text{Hip}_1 \text{Hip}_2$.

Using our method of substitution, we get, step by step:

$$\models [10] \supset \text{Hip} \vdash_0 \supset D e_{11c} f_{11c} g_{11c} h_{11c} i_{11c} k_{11c} (2) x_{11c} y_{11c} z_{11c}$$

$D_2 x_{11c} y_{11c} z_{11c} \text{ is a theorem.}$

Then we have by the direction of *modus ponens* of $\vdash_0 0$,

$$\models [10] \supset \text{Hip} \supset \vdash_0 D e_{11c} f_{11c} g_{11c} h_{11c} i_{11c} k_{11c} (2) x_{11c} y_{11c} z_{11c}$$

$\vdash_0 D_2 x_{11c} y_{11c} z_{11c} \text{ is a theorem.}$

Now, using the logical calculus of $\models [10] c$, we get:

$$\models [10] \supset S_0 D e_{1c} f_{1c} g_{1c} h_{1c} i_{1c} k_{1c} (0) x_{1c} y_{1c} z_{1c} S'_0 (D)_0 x_{1c} y_{1c} z_{1c} \text{ is a theorem.}$$

Then we get by (5),

$$\models [10] \supset D e_{1c} f_{1c} g_{1c} h_{1c} i_{1c} k_{1c} (0) x_{1c} y_{1c} z_{1c} S'_0 (D)_0 x_{1c} y_{1c} z_{1c} \text{ is a theorem.}$$

This theorem enables us to prove by the logical calculus of $\models [10] c$,

$$(12) \quad \models [10] \supset D_0 x_{1c} y_{1c} z_{1c} S'_0 (D)_0 x_{1c} y_{1c} z_{1c} \text{ is a theorem.}$$

The same method can be applied to $\text{Ax}_0 Z$, yielding:

$$(13) \quad \models [10] \supset \text{Ax}_0 z_{1c} S'_0 (\text{Ax})_0 z_{1c} \text{ is a theorem.}$$

The calculus of $\vdash_0 0$ enables us to prove:

$$\models [10] \vdash_0 \supset Ax_2 u_{31} \vdash_2 u_{31} \text{ is a theorem.}$$

Then we get, by the method of substitution used above,

$$\models [10] \supset Im_1 z_{1c} z_{11c} \vdash_0 \supset Ax_2 z_{11c} \vdash_2 z_{11c} \text{ is a theorem.}$$

Now, using the direction of *modus ponens* of $\vdash_0 0$, we get:

$$\models [10] \supset Im_1 z_{1c} z_{11c} \supset \vdash_0 Ax_2 z_{11c} \vdash_0 \vdash_2 z_{11c} \text{ is a theorem.}$$

Then we get by the logical calculus of $\models [10] c$,

$$\models [10] \supset S'_0 (Ax)_0 z_{1c} S_{02} z_{1c} \text{ is a theorem.}$$

Then we have by (13),

$$(14) \quad \models [10] \supset Ax_0 z_{1c} S_{02} z_{1c} \text{ is a theorem.}$$

Now let us remark that we have by the calculus of $\vdash_0 0$:

$$\models [10] \vdash_0 \supset \vdash_2 x_{31} \supset \vdash_2 y_{31} \supset D_2 x_{31} y_{31} z_{31} \vdash_2 z_{31} \text{ is a theorem.}$$

Then we have by the method of substitution used above:

$$\models [10] \supset Hip_1 \vdash_0 \supset \vdash_2 x_{11c} \supset \vdash_2 y_{11c} \supset D_2 x_{11c} y_{11c} z_{11c} \vdash_2 z_{11c} \text{ is a theorem.}$$

Applying the direction of *modus ponens* to $\vdash_0 0$ three times to this theorem, we get:

$$\models [10] \supset Hip_1 \supset \vdash_0 \vdash_2 x_{11c} \supset \vdash_0 \vdash_2 y_{11c} \supset \vdash_0 \vdash_2 D_2 x_{11c} y_{11c} z_{11c} \vdash_0 \vdash_2 z_{11c} \text{ is a theorem.}$$

Then we get by the calculus of $\models [10] c$ and by (12),

$$(15) \quad \models [10] \supset \wedge S_{02} x_{1c} \wedge S_{02} y_{1c} D_0 x_{1c} y_{1c} z_{1c} S_{02} z_{1c} \text{ is a theorem.}$$

From (14) and (15) we get by Hetper's method:

$$(16) \quad \models [10] \supset \vdash_0 u_{1c} S_{02} u_{1c} \text{ is a theorem.}$$

This is our fundamental lemma.

In an analogous manner we get, using $\vdash_0 0$ instead of $\models [10] c$:

$$(17) \quad \models [10] \vdash_0 \supset \vdash_2 u_{31} S_{24} u_{31} \text{ is a theorem.}$$

The theorems (16), (17) enable us to get Gödel's theorem by an easy formal derivation.

V. Let us introduce the following directions of abbreviation:

$G_L(E)$ is an abbreviation of $\exists [.0(L) L] \bar{x}_L \bar{y}_L \bar{z}_L \bar{u}_L$

$$\wedge = E \Pi [.2(L) .1(L)] \bar{y}_L \bar{z}_L [.0(L)]$$

$$\wedge Im_{.0(L)} E \bar{x}_L \wedge (\bar{z}_L \bar{y}_L \bar{x}_L \bar{u}_L) [.0(L)] \sim \vdash_L \bar{u}_L.$$

$G_{(L)}$ is an abbreviation of $G_L(\Pi [.2(L) .1(L)] \bar{a}_{.1(L)} G_{.1(L)}(\bar{a}_{.1(L)}))$.

Now we have by elementary logical calculus:

(18) $\models [1\ 0] \vdash_0 \equiv G_{(2)} \sim \vdash_2 G_{(4)}$ *is a theorem.*

(19) $\models [1\ 0] \vdash_0 \vdash_2 \equiv G_{(4)} \sim \vdash_4 G_{(6)}$ *is a theorem.*

On the other hand we have by (16) and (17),

(20) $\models [1\ 0] \supset \vdash_0 G_{(2)} \vdash_0 \vdash_2 G_{(4)}$ *is a theorem,*

(21) $\models [1\ 0] \supset \vdash_0 \sim G_{(2)} \vdash_0 \vdash_2 \sim G_{(4)}$ *is a theorem,*

and,

(22) $\models [1\ 0] \vdash_0 \supset \vdash_2 G_{(4)} \vdash_2 \vdash_4 G_{(6)}$ *is a theorem,*

(23) $\models [1\ 0] \vdash_0 \supset \vdash_2 \sim G_{(4)} \vdash_2 \vdash_4 \sim G_{(6)}$ *is a theorem.*

From (18) we get by the logical calculus of $\vdash_0$:

(24) $\models [1\ 0] \supset \vdash_0 G_{(2)} \vdash_0 \sim \vdash_2 G_{(4)}$ *is a theorem.*

(25) $\models [1\ 0] \supset \vdash_0 \sim G_{(2)} \vdash_0 \vdash_2 G_{(4)}$ *is a theorem.*

Now we get from (20), (24),

(26) $\models [1\ 0] \supset \vdash_0 G_{(2)} \vdash_0 = .0\ (3) .1\ (3)\ [3]$ *is a theorem.*

From (21), (25) we get:

$\models [1\ 0] \supset \vdash_0 \sim G_{(2)} \vdash_0 \vdash_2 = .0\ (5) .1\ (5)\ [5]$ *is a theorem.*

Therefore, since by (16) and (26),

$\models [1\ 0] \supset \vdash_0 G_{(2)} \vdash_0 \vdash_2 = .0\ (5) .1\ (5)\ [5]$ *is a theorem,*

we have that

$\models [1\ 0] \supset \forall \vdash_0 G_{(2)} \vdash_0 \sim G_{(2)} \vdash_0 \vdash_2 = .0\ (5) .1\ (5)\ [5]$ *is a theorem.*

This theorem corresponds closely to the first theorem of Gödel. Notice, however, that there the contradiction does not appear in $\vdash_0 0$ but in $\vdash_2 2$.

Now we pass to the second theorem of Gödel. From the theorems (22) and (19) we get:

$\models [1\ 0] \vdash_0 \supset \vdash_2 G_{(4)} \vdash_2 \sim G_{(4)}$ *is a theorem.*

Thus we have:

$\models [1\ 0] \vdash_0 \supset \sim G_{(2)} \vdash_2 = .0\ (5) .1\ (5)\ [5]$ *is a theorem,*

$\models [1\ 0] \vdash_0 \supset \sim \vdash_2 = .0\ (5) .1\ (5)\ [5]$ $G_{(2)}$ *is a theorem.*

Now, applying *modus ponens* of $\vdash_0 0$ and using (26), we get:

$\models [1\ 0] \supset \vdash_0 \sim \vdash_2 = .0\ (5) .1\ (5)\ [5] \vdash_0 = .0\ (3) .1\ (3)\ [3]$ *is a thm.*

We see that, if we could prove in $\vdash_0 0$ that there is no contradiction in $\vdash_2 2$, we should have a contradiction in $\vdash_0 0$. This is Gödel's theorem.

Let us remark that there is no way to prove:

If $\models [1\ 0] \sim \vdash_0 (3\ 2) = .0\ (3) .1\ (3)\ [3]$ is a theorem, then

$\models [1\ 0] \vdash_0 (3\ 2) \sim \vdash_2 (5\ 4) = .0\ (5) .1\ (5)\ [5]$ *is a theorem.*

Nevertheless it is obvious that if we could prove,

$$\models [10] \sim \vdash_0 (32) = .0(3).1(3)[3] \text{ is a theorem,}$$

we could prove applying the same method step by step, that

$$\models [10] \vdash_0 (32) \sim \vdash_2 (54) = .0(5).1(5)[5] \text{ is a theorem.}$$

Thus there is no hope of proving the consistency of $\vdash_0 (32) 0$.

VI. Summary of the directions concerning the use of variables in the systems and metasystems occurring in this paper:

Systems and metasystems	Constant expressions	Real variables	Apparent variables	Domain of substitution
The verbal system	any expression	$E, F, G, H, I, K, L, M, N$	—————	any expression
The auxiliary system	any expression of type c	—————	—————	—————
The system $\models [10]c$	any expression of type 1	$a_{1c}, b_{1c}, c_{1c},$	$\bar{a}_0, \bar{b}_0, \bar{c}_0,$	$.0(1),$ $* a_{1c} b_{1c},$
		$x_{1c}, y_{1c}, z_{1c},$ $a_{11c}, b_{11c}, c_{11c},$	$\bar{x}_0, \bar{y}_0, \bar{z}_0,$ $\bar{a}_{10}, \bar{b}_{10}, \bar{c}_{10},$	$* .0(1) * a_{1c} b_{1c},$ $* * a_{1c} b_{1c} .0(1),$
		$x_{11c}, y_{11c}, z_{11c}$	$\bar{x}_{10}, \bar{y}_{10}, \bar{z}_{10}$	
The metasystem $\vdash_0 (32) 0$	any expression of type 3	$a_{31}, b_{31}, c_{31},$	$\bar{a}_2, \bar{b}_2, \bar{c}_2,$	$.0(3),$ $* a_{31} b_{31},$
		$x_{31}, y_{31}, z_{31},$ $a_{131}, b_{131}, c_{131},$	$\bar{x}_2, \bar{y}_2, \bar{z}_2,$ $\bar{a}_{12}, \bar{b}_{12}, \bar{c}_{12},$	$* .0(3) * a_{31} b_{31},$ $* * a_{31} b_{31} .0(3),$
		$x_{131}, y_{131}, z_{131}$	$\bar{x}_{12}, \bar{y}_{12}, \bar{z}_{12}$	
The metasystem $\vdash_2 (54) 2$	any expression of type 5	$a_{53}, b_{53}, c_{53},$	$\bar{a}_4, \bar{b}_4, \bar{c}_4,$	$.0(5),$ $* a_{53} b_{53},$
		$x_{53}, y_{53}, z_{53},$ $a_{153}, b_{153}, c_{153},$	$\bar{x}_4, \bar{y}_4, \bar{z}_4,$ $\bar{a}_{14}, \bar{b}_{14}, \bar{c}_{14},$	$* .0(5) * a_{53} b_{53},$ $* * a_{53} b_{53} .0(5),$
		$x_{153}, y_{153}, z_{153}$	$\bar{x}_{14}, \bar{y}_{14}, \bar{z}_{14}$	

Note that in X_{KL} , $\bar{X}_K$, X_{IKL} , and $\bar{X}_{IL}$, K and L stand for types, while I is used to build up new letters.⁶

INSTYTUT LOGIKI, LWÓW

⁶ Cf. *New foundation*, A₁₁ and p. 35.